

T-OCS V7.0

REVISION-2

NON-CONFIDENTIAL CORPORATE TECHNICAL & TECHNICAL-QUALIFICATION PRESENTATION

A technical discussion designed to determine whether the engineering hypothesis merits a bounded evaluation program.

Meeting outcome

Technical scope • required evidence • entry criteria
• NDA • next action

Engineering stance

Hypothesis, not product claim. Evidence before
commitment

L1 NON-CONFIDENTIAL

NO PERFORMANCE CLAIMS

CONTROLLED DISCLOSURE

John builds • Peter red-teams • Naif decides

What this meeting must produce

The meeting is successful only if the next engineering step becomes measurable.

Technical Scope

What exact work can the organization perform?

Required Evidence

What must be available before that work starts?

Entry Criteria

What conditions must be true for Phase-1?

NDA + Commercial

What can be shared, and under what commercial structure?

Next Action

Who does what, by when, with what evidence?

1

Close the loop

2

Meeting result
→ Engineering action
→ Deliverable
→ Acceptance criterion
→ Owner
→ Deadline

3

4

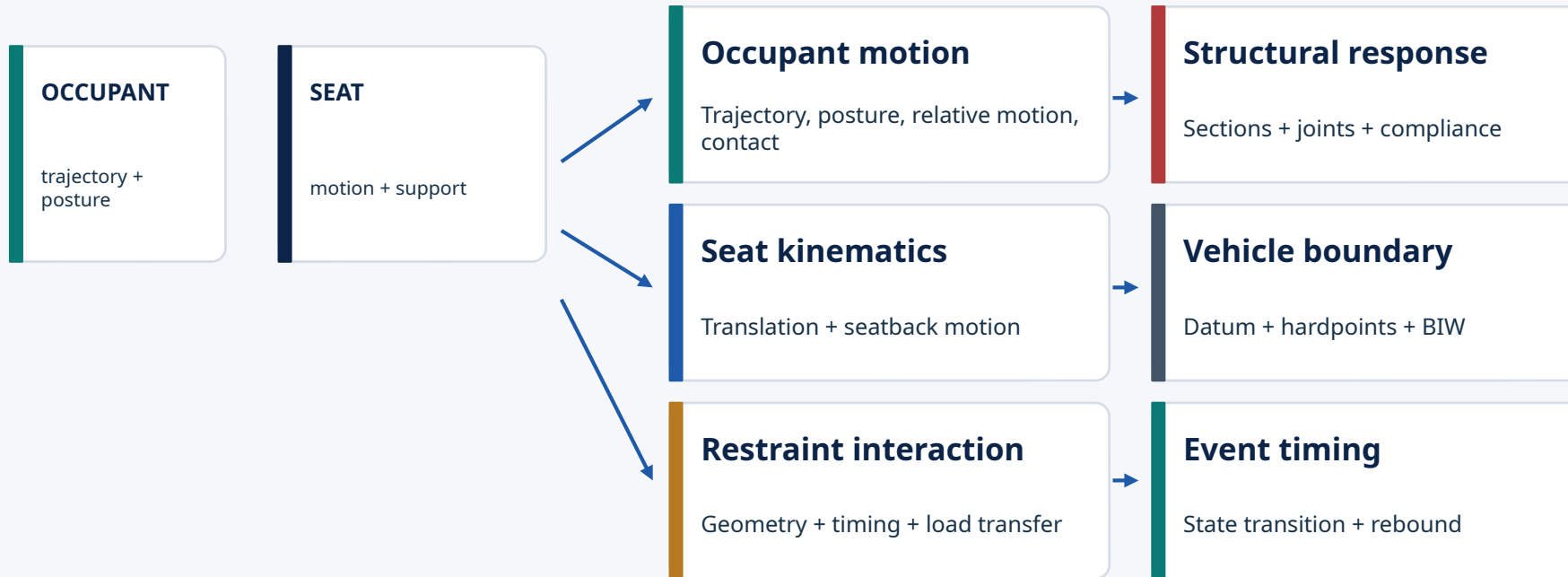
Hard stop

No unresolved technical thread becomes a hidden commitment.

5

The problem is coupled — not a single “rear-seat motion” metric.

T-OCS studies how seat, occupant, restraint, structure and vehicle boundary interact in a dynamic event.



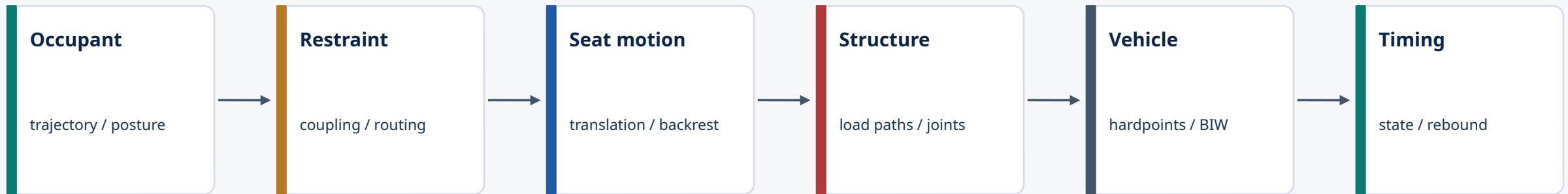
Engineering question

Can these interactions be coordinated and measured without creating new structural, packaging, restraint or timing conflicts?

The system behaves as a chain of coupled boundary conditions.

A change in one block can alter the load or motion seen by another.

Coupling logic



Local optimization can move the problem

More stiffness may reduce motion but increase transmitted load. More travel may help time/stroke but create package/end-stop demands.

Interface changes propagate

Moving an interface can change packaging, restraint geometry and structural load transfer simultaneously.

System evidence is required

Therefore: controlled engineering → measurement → testing → correlation.

A feature-level gain can become a system-level trade-off.

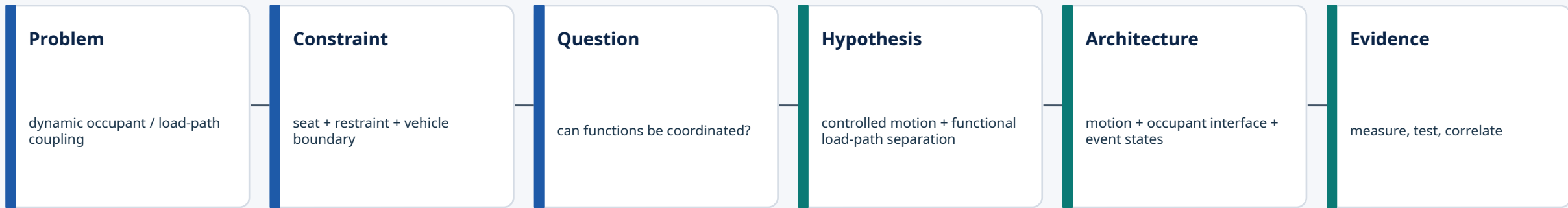
The unit of evaluation is the coupled seat + occupant + restraint + vehicle boundary.

Increase controlled motion Changes stopping-time/stroke relationship	System interaction Seat/occupant trajectory + structural reactions	Required evidence Configuration-controlled measurement and correlation
Increase stiffness May reduce motion but raise transmitted load	System interaction Force history + joint/section response	Required evidence Configuration-controlled measurement and correlation
Move an interface May clear one region but change another	System interaction Envelope + restraint + load-path check	Required evidence Configuration-controlled measurement and correlation
Change restraint routing May alter occupant trajectory	System interaction Timing + geometry + occupant motion	Required evidence Configuration-controlled measurement and correlation

Engineering conclusion: optimize the system response, not one isolated feature.

How the engineering hypothesis is structured

This is a logic reconstruction, not a claimed historical invention chronology.

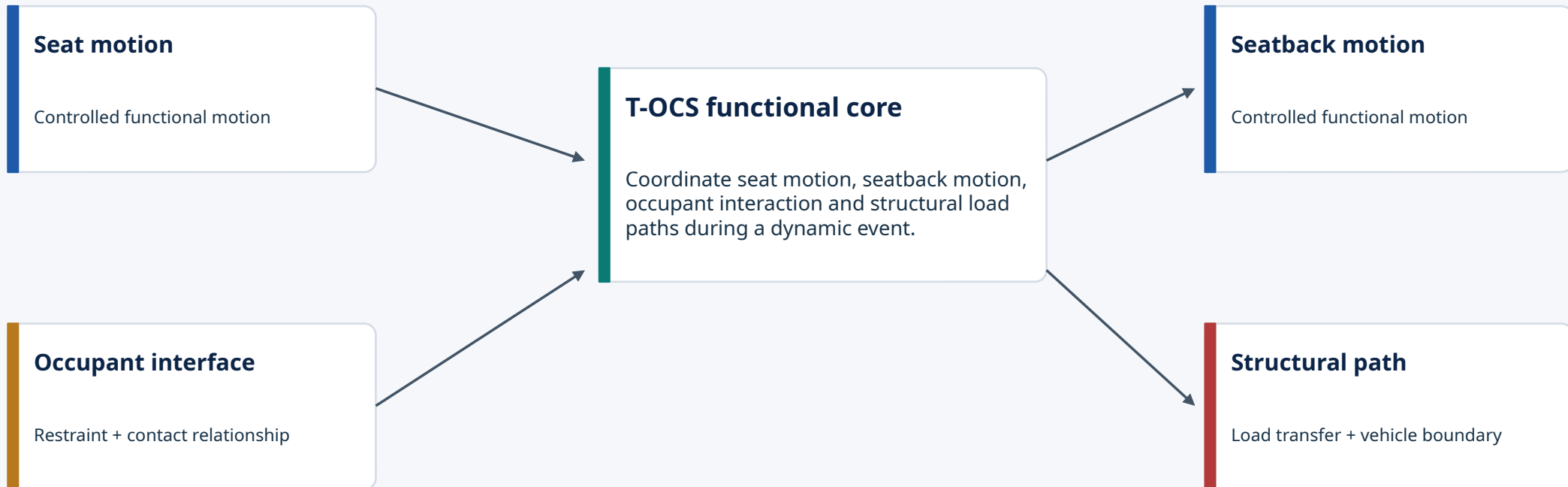


HISTORICAL CHRONOLOGY: NOT ESTABLISHED

Where the controlling source does not document why a specific feature was first conceived, the correct label is NOT ESTABLISHED — SOURCE REQUIRED.

At L1, T-OCS is a coordinated seat-system hypothesis.

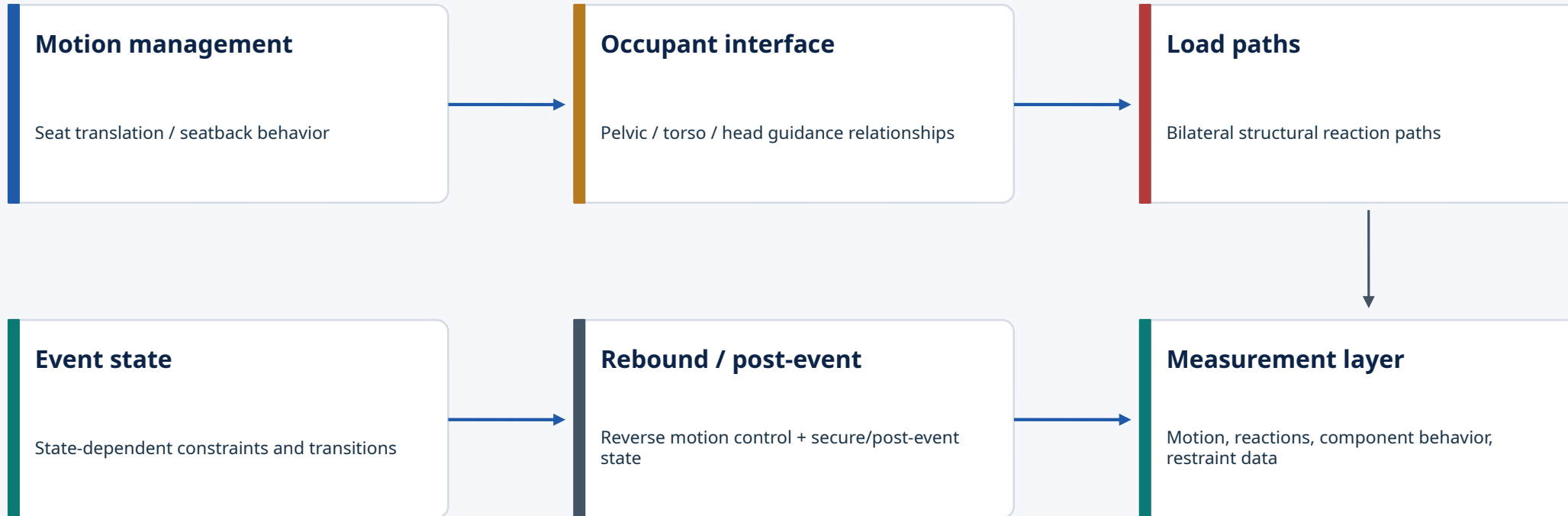
High-level function can be understood without implementation detail.



L1 message: the value is in the coordinated system hypothesis — not an unproven performance claim.

The architecture is expressed as functions and evidence links.

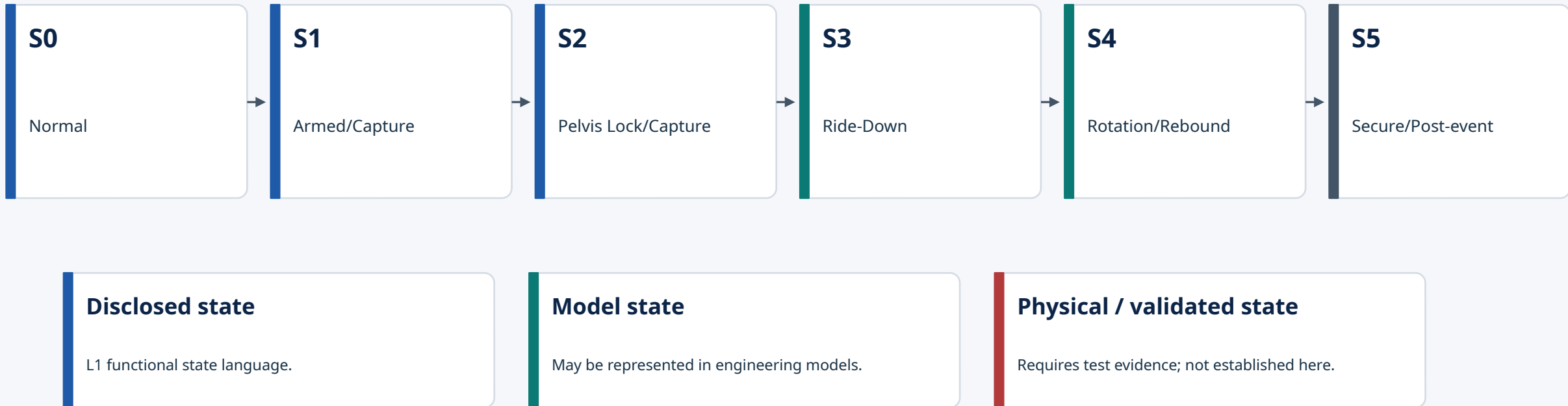
Functional blocks can be evaluated independently, but their system interactions must remain visible.



Every functional block ultimately needs a measurable evidence path.

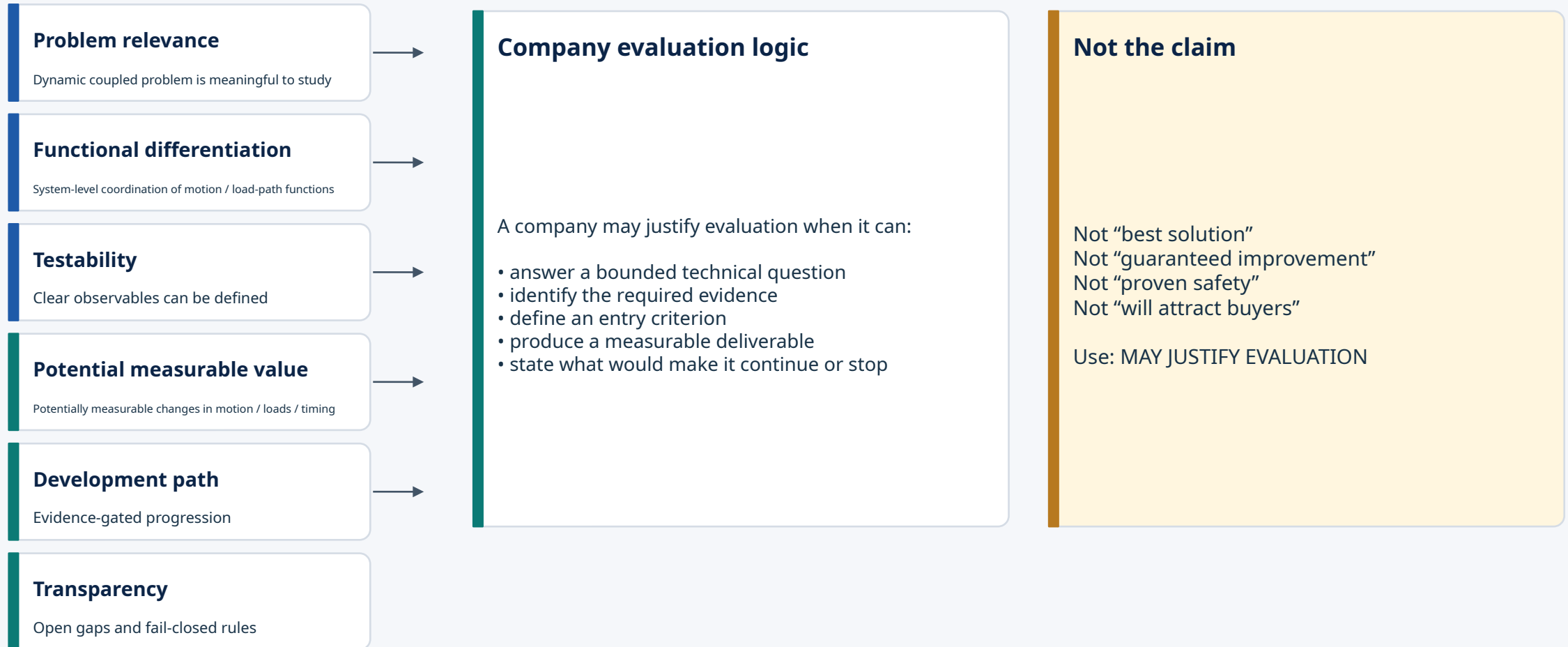
Canonical external state vocabulary

States are a control language — not a statement that the physical transition behavior has been validated.



The evaluation case is evidence-oriented, not superiority-oriented.

The question is whether the architecture creates a credible engineering opportunity worth bounded investigation.



Observables are targets for evidence — not current T-OCS results.

A useful Phase-1 discussion starts with measurable outputs rather than marketing metrics.

Seat motion

displacement • velocity • acceleration

Evidence status

Target observable — NOT a current result

Seatback

angle • angular rate / acceleration

Evidence status

Target observable — NOT a current result

Structure

reactions • load transfer • asymmetry

Evidence status

Target observable — NOT a current result

Occupant

trajectory • relative motion • contact timing

Evidence status

Target observable — NOT a current result

Restraint

timing • load history • geometry

Evidence status

Target observable — NOT a current result

Components

force-displacement • energy • rebound behavior

Evidence status

Target observable — NOT a current result

Controlled reconstruction baseline — not a product release.

The current evidence base supports engineering discussion, not production or safety conclusions.

Controlled

Functional architecture

Controlled

Engineering intent / requirements framework

Controlled

Controlled reconstruction / model artifacts

Controlled

Evidence and change-control rules

Not yet proven

Physical validation

Not yet proven

Final manufacturability

Not yet proven

Final material / fastener closure

Not yet proven

Absorber / lock / rebound validation

Not yet proven

Production readiness

Current V7 classification: research / engineering reconstruction baseline.

Evidence classes are kept separate.

This prevents source studies, design intent and model outputs from becoming “T-OCS results” by accident.

Source-derived V7/V6 disclosure material	Support Problem / architecture framing	Classification Do not promote beyond this class
Design intent Functional decomposition + controlled interfaces	Support What the system is intended to do	Classification Do not promote beyond this class
Model-level Controlled reconstruction / kinematic representations	Support Model behavior only	Classification Do not promote beyond this class
Physical evidence Not established as complete T-OCS validation	Support Cannot support current performance claims	Classification Do not promote beyond this class
Unknown / open Vehicle inputs, material/joint closure, test behavior, production evidence	Support Defines next evidence gates	Classification Do not promote beyond this class

The blockers are evidence gaps, not reasons to invent values.

The next move is evidence closure and independent verification, not cosmetic certainty.

Vehicle integration

Datum • H/R • hardpoints • BIW • restraint anchors

Engineering definition

Released dimensions • PMI/GD&T • tolerances

Physical components

Materials • fasteners • joints • absorber • lock • rebound

Validation

Measurements • dynamic tests • correlation • vehicle evidence

Fail-closed rule

Missing = UNKNOWN / OPEN / NOT ESTABLISHED. Conflicting values are preserved, not reconciled by convenience.

Every transition is an evidence gate.

Later maturity is never inferred from earlier artifacts.



PROPOSED DEVELOPMENT PATH — not a claim that every gate has been executed.

Gate discipline

Before prototype: fabrication definition, materials/joints, fixture and inspection controls. Before testing: article + instrumentation + test case + acceptance criteria.

Where a company can enter the program

Entry is through a bounded technical problem, not a blanket commitment to “finish V7.”

Strategic Engineering Partner

Requirements / CAD / CAE / feasibility

Test Laboratory

Dynamic test / instrumentation / measurement

Seat Engineering

Advanced development / technical fit

Prototype / Manufacturing

Build definition / process development

Research / Sponsorship / Investment

Program support subject to explicit agreement

INTEREST $\neq$ COMMITMENT

Commercial structure is a question for the counterpart, not a project assumption.

Ask what the organization actually offers for a bounded engineering result.

Paid Engineering

Fixed deliverable + acceptance

Co-development

Shared scope / governance if explicitly agreed

Test Services

Quoted test + data outputs

Research

Defined research question + evidence

Sponsorship / Investment

Separate commercial decision; no implied commitment

Commercial firewall

No “interested” → partner; no discussion → contract; no proposal → commitment.

Minimum credible engineering package — bounded, acceptance-controlled.

Phase-1 is not “finish V7” and not “close the entire gap universe.”

Scope

Feasibility, package, interface, kinematic and evidence-plan review

Inputs

L1 brief + approved engineering context + agreed questions

Work

Assess → identify issues → define evidence → propose next actions

Outputs

Technical assessment + issue/evidence register + next-decision package

Acceptance

Known/open/unknown clearly separated; owner + evidence + deadline captured

Out of scope

Production release, crash validation, regulatory certification, blanket V7 closure

Commercial form: one bounded result with defined acceptance. Hours are not the purchased output; price is organization-specific and TBD.

Use the meeting to extract decisions, not compliments.

Every C closes to a concrete record.

C1 Technical Scope

What can you do?

C2 Required Evidence

What do you need?

C3 Entry Criteria

What must be true to start?

C4 Commercial Structure

How would the bounded work be procured?

C5 NDA Trigger

What can be shared now vs under NDA?

Required meeting close

Technical Scope → Evidence → Entry → Commercial → NDA → Action → Deliverable → Acceptance → Owner → Deadline

Three different entry points, one discipline: extract the actual scope.

Current outreach records are used only to establish meeting context — not partnership status.

ARRK Engineering

Passive Safety → CAD → CAE → Prototype → Acceptance

Contact: Claudius Brix — VP / Passive Safety

Scheduling: Microsoft Teams proposed; date/time pending

Calspan Automotive Test Services

Dynamic Test → Instrumentation → Measurement → Correlation

Contact: William McGowan — Business Unit Director

Scheduling: Afternoon ET preference; date/time pending

Toyota Boshoku America

Seat Advanced Development → Technical Fit → Co-development

Contact: Jacob Richter — Senior Engineer, Advanced Development

Scheduling: Sep 23 or 24, 2026 • 11:00 AM ET; L1 requested

Commitment firewall

Meeting response = source record. It does not equal partner, customer, sponsor, investor, or approved test-provider status.

Answer discipline: KNOWN → LIMITATION → EVIDENCE NEEDED → NEXT ACTION

Do not defend uncertainty with rhetoric.

What is proven? Architecture + controlled definitions + model-level artifacts.	LIMITATION Physical performance still open.	NEXT ACTION Scope the evidence gate.
Do you have native CAD? Current external package is a controlled engineering/reconstruction baseline.	LIMITATION Do not describe source format or release status beyond the approved disclosure boundary.	NEXT ACTION Scope the evidence gate.
Do you have physical test data? Complete T-OCS physical validation is not established.	LIMITATION Test article + instrumentation + raw data required.	NEXT ACTION Scope the evidence gate.
Why evaluate? Testable system hypothesis + bounded evidence path.	LIMITATION Not a superiority claim.	NEXT ACTION Scope the evidence gate.
What would stop the program? Phase-1 falsification criteria.	LIMITATION Agree stop criteria before deeper spend.	NEXT ACTION Scope the evidence gate.

The program should actively try to break its own hypothesis.

Credibility improves when the project can state what would make it stop.

Could packaging invalidate it?

Test with vehicle-specific envelope and interfaces.

Could kinematics become unworkable?

Test joint / end-stop / interference feasibility.

Could load paths become asymmetric?

Measure bilateral reactions / structural response.

Could component behavior be unsuitable?

Characterize force / displacement / rebound behavior.

Could test data fail to correlate?

Define comparable observables + configuration controls.

Falsification logic

Define the minimum discriminating evidence before full-scale spend.

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Question to the counterpart: “What is the cheapest credible test you would trust to challenge this hypothesis?”

Turn disagreements into evidence decisions.

Use the same conversion grammar for requirements, packaging, kinematics, loads and testability.



Example topics

Requirement conflict • Packaging/BIW interface • Kinematic feasibility • Load/energy uncertainty • Testability/correlation

Ideas are welcome. Uncontrolled baseline changes are not.

Every idea receives a classification before it becomes an engineering action.

A Compatible Improvement

Fits current Design Intent / boundary

B Architecture Modification

Changes topology / function

C Design-Intent Change

Changes protected intent / requirements

D IP / Disclosure Conflict

Requires stop + escalation

Idea → Technical Impact → Evidence → Decision → Owner → Deadline

NO AUTOMATIC V7 CHANGE

One task = one accountable owner = one deadline.

Meeting quality depends on converting conversation into traceable execution.

Requirements		Owner: _____	Deadline: _____	Evidence: _____
CAD		Owner: _____	Deadline: _____	Evidence: _____
CAE / FE		Owner: _____	Deadline: _____	Evidence: _____
Prototype		Owner: _____	Deadline: _____	Evidence: _____
Testing		Owner: _____	Deadline: _____	Evidence: _____
Instrumentation		Owner: _____	Deadline: _____	Evidence: _____
Measurement		Owner: _____	Deadline: _____	Evidence: _____
Correlation		Owner: _____	Deadline: _____	Evidence: _____
Evidence		Owner: _____	Deadline: _____	Evidence: _____
IP / Disclosure		Owner: _____	Deadline: _____	Evidence: _____
Commercial / Program		Owner: _____	Deadline: _____	Evidence: _____

Do not ask the company to trust an unproven claim.

Ask the company to judge whether the engineering program is controlled enough to test.

Clear problem

The engineering question is explicit

Visible gaps

Unknowns are named

Evidence gates

Each maturity step has a proof requirement

Independent review

Peter red-teams before external release

Bounded milestones

Phase-1 has acceptance criteria

Falsification

The program can stop if the hypothesis fails

Credibility statement

“We are not asking you to believe that the technology works. We are asking whether the problem, the engineering hypothesis, the evidence path, and the bounded Phase-1 are worth evaluating.”

Do not end with “Thank you.” End with a measurable next step.

Complete the fields live before leaving the meeting.

Technical Scope

Required Evidence

Entry Criteria

NDA Status

Next Action

Owner

Deadline

Next Review

Close rule

No hidden commitment. No open-ended scope. Read back owner + deadline + evidence before adjournment.